

WLA4ND: a Wearable Dataset of Learning Activities for Young Adults with Neurodiversity to Provide Support in Education

Hui Zheng
Click Therapeutics
New York, USA
hzheng5@gmu.edu

Pattiya Mahapasuthanon
George Mason University
Fairfax, VA, USA
pmahapas@gmu.edu

Yujing Chen
George Mason University
Fairfax, VA, USA
ychen37@gmu.edu

Huzefa Rangwala
George Mason University
Fairfax, VA, USA
rangwala@gmu.edu

Anya S. Evmenova
George Mason University
Fairfax, VA, USA
aevmenov@gmu.edu

Vivian Genaro Motti
George Mason University
Fairfax, VA, USA
vmotti@gmu.edu

ABSTRACT

Data-driven assistive wearable technologies are promising to support young adults with neurodiversity in inclusive education. Existing datasets focus on wearable sensor data of various activities from neurotypical people. However, no dataset exists including learning-related activity data from individuals with neurodiversity. The contributions of this paper include (1) WLA4ND, a dataset of learning activities performed by eight young adults with neurodiversity collected from smartwatch sensors. The activities are common learning tasks, including reading, writing, typing, answering follow-up questions, and off-task. (2) Evaluation of classification on WLA4ND with five activity recognition models, including conventional and deep learning methods. The Convolutional Recurrent Neural Network (CRNN) model achieved a balanced accuracy of 92.2% for user-dependent evaluations, while Federated Multi-Task Hierarchical Attention Model (FATHOM) achieved 91.8% for user-independent evaluations. This evaluation demonstrates that existing activity recognition technologies can be applied to neurodiverse populations. Also, WLA4ND can be used by researchers as a complement for activity recognition, automatic labeling, and next-generation assistive wearable applications.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility**; *Empirical studies in accessibility*; *Ubiquitous and mobile computing*.

KEYWORDS

Datasets, Neurodiversity, Wearable Sensors, Smartwatches, Ubiquitous Computing, Inclusive Education, Classification

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1 INTRODUCTION

Neurodiverse conditions consist of variations in the abilities of individuals in cognitive, motor, and communication skills [19, 56, 57]. Neurodiverse conditions also impact the individual's behaviors and memory, which may impair their executive functions [27, 41]. Approximately 17% of the children in the United States present neurodiverse conditions [18]. Those conditions vary in severity level per individual [40], however they are lifelong and chronic. They also affect individuals' opportunities for employment, education, and independent living. Conversely, neurodiverse conditions have been associated with positive traits as well, including pattern recognition, hyperfocus, and creativity [1]. Because the common traits in neurodiversity affect the individual autonomy in daily activities, support from a caregiver or an assistant is often required [6, 72]. In-person monitoring and interventions by caregivers, aides, or assistants are effective. However, they are limited regarding consistency, accountability, and scalability [72]. Moreover, verbal interventions to moderate behavior may stigmatize and cause discomfort to a neurodiverse individual by explicitly signaling a disability [43]. Verbal interventions may also be error-prone and subjective due to the assistants' fatigue or bias [60].

To facilitate access and improve opportunities to inclusive higher education programs for young adults with neurodiversity, in 2008 the Higher Education Opportunity Act (HEOA) was introduced in the United States. Despite HEOA increasing the education opportunities for adults with neurodiversity, the variations on cognitive skills and adaptive functioning still affect multiple aspects of their lives, such as opportunities for employment, independent living, and engagement in traditional learning activities [47]. Conventionally, in inclusive settings of special/inclusive education programs, support staff is hired to provide continuous assistance for students with neurodiversity. Nonetheless, the support staff can not always be near the students with neurodiversity, for instance, when the students are off-campus or taking online classes. This challenge became particularly important during the COVID-19 pandemic since most academic courses switched to online learning environments,

and personal support staff is absent or remote when students with neurodiversity are completing their learning activities.

To address such shortcomings, there is a promising potential to further explore ubiquitous computing, including mobile and wearable applications that deliver timely interventions and reminders on demand. The possible options for personalization and automation make assistive wearables suitable to meet the needs of users with neurodiversity, especially individuals who need continuous assistance with executive functioning [16, 60]. Specifically, the monitoring of physiological and motion data from wearable sensors yields promising opportunities for wearable assistive technologies [13]. With the data collected from wearable sensors, assistive wearable applications have a great potential to monitor users' activities and provide strategic interventions. Additionally, the involvement of users with neurodiversity is important in the iterative design process to identify requirements and to design suitable assistive applications [20, 56, 58]. Assistive wearable applications can help neurodiverse users with self-regulation, monitoring their behaviors to inform them and raise their awareness, and also to mediate the assistance they receive from caregivers [43]. Although assistive wearables are well-suited to provide support, no inclusive dataset of learning activities involving neurodiverse users exist.

To fully realize the potential wearable applications to facilitate assistance, it is essential to have a dataset of users' activities subject to monitoring and intervention. For validity, such data has to be collected from users with neurodiversity, ensuring that activity recognition models match end users' profiles [70]. Given the unique abilities and motor skills of neurodiverse users, a learning activity dataset of individuals with neurodiversity collected from wearable sensors complements datasets of activity recognition for neurotypical people. A dataset for neurodiverse users facilitates assistive wearable technologies driven by data and machine learning models, enabling monitoring of students as well as the delivery of less intrusive interventions without requiring in-person assistance. Datasets and activity recognition models pave the way to provide remote support for students with neurodiversity.

This paper presents WLA4ND, a wearable sensor dataset of learning activities. We designed four learning activities which were performed by eight participants, resulting in a total of 32 learning sessions. The eight young adults who participated in the study are neurodiverse, including conditions such as Autism Spectrum Disorders (ASD), Attention Deficit and Hyperactivity Disorder (ADHD), and Intellectual and Developmental Disability (IDD). The learning activities are hand activities (reading with follow-up questions, writing, and typing) which are commonly performed in inclusive education. We used built-in smartwatch sensors (accelerometer, gyroscope, and heart rate sensors) to collect data. The dataset was carefully annotated by three coders following a multi-label approach. The dataset includes six ground truth labels: (1) Read, (2) Write Q&A, (3) Write, (4) Type, (5) Off-task, and (6) Rest. Furthermore, we investigated five state-of-the-art classification approaches and evaluated them on this novel WLA4ND dataset, to explore whether and to what extent the existing activity recognition technologies suit data collected from individuals with neurodiversity.

The contributions of this paper include: (1) WLA4ND, a novel dataset of learning activities performed by young adults with neurodiversity collected via wearable sensors; (2) investigation of classification-based activity recognition models evaluated on this novel WLA4ND dataset. Having high accuracy on classification tasks from activity recognition models accelerates the automation of assistance, creating promising opportunities for applications of AI-based assistive wearable technologies in inclusive education. Ultimately, such contributions will lead to improvement in the quality of life for individuals with neurodiversity.

2 RELATED WORK

Assistive technologies (ATs) have great potential to meet users' needs, monitoring their activities and providing them with support on demand. Assistance can be delivered through ubiquitous, mobile, and wearable technologies, and as such reducing potential stigmas, promoting and improving individuals' independence [15]. ATs can augment, increase, maintain, or improve the functional capabilities of persons with disabilities. This type of support is beneficial across user groups regardless of their specific needs. Still, prior work indicates that most assistive technologies that employ artificial intelligence (AI) (including applications of natural language processing [36] and computer vision [35]) are created for users with visual, hearing, speech, and motor impairments. Meanwhile, the ATs for users with neurodiversity are limited. Most technologies have also been developed for children [49]. To provide an overview of the latest research in this domain, in this section, we focus on three aspects: (1) ATs to assist people with neurodiversity in inclusive education and independent living, (2) wearable ATs developed to provide support for people with neurodiversity, and (3) wearable or mobile sensor datasets related to human activities.

2.1 Assistive Technologies for Neurodiverse Users in Inclusive Education and Independent Living

ATs designed for people with neurodiversity focus on providing educational support, enhancing communication strategies and skill development, and preparing individuals to live independently. Prior work about ATs include studies about learning games [7, 33], augmentative and alternative communication (AAC) technologies [23, 50, 68], playground equipment [69, 71], multimedia technologies [5], and virtual reality (VR) technologies [52].

To improve the engagement and academic skills of students with diverse needs, collaborative Kinect games were used in classroom settings. Such games captured and augmented motion-based activities from students [7, 33]. Apart from education, several ATs have been studied to aid with social and communication skills. For instance, interactive balls [69, 71], Kinect games [39], and tablet applications [68] were used to encourage social interaction and social bonds among children with ASD.

With ATs focused on improving cognitive and academic skills, communication, and social interactions, researchers have also explored ATs to teach living skills to neurodiverse users. Video contents were created with a 360° camera allowing users to wear a head-mounted device to experience common activities of daily living and acquire related skills, such as shopping skills, commuting skills, and vocational skills [52]. Additionally, social media and video content were used to support young adults with IDD to further develop their social skills [5]. An interactive game including

Table 1: A comparative analysis of the sample sizes, activities, classification models, and results from existing datasets reported in the Related Work section.

Dataset	Sample Size	Activity Types	Classification Model	User-Dependent	User-Independent
ExtraSensory [63]	60	Fitness & Behavior	MLP	-	77.3% accuracy
Laput & Harrison [34]	50	Hand	CRNN	90.7% accuracy	79.2% accuracy
Weiss et al. [67]	17	Fitness & Hand	Random Forest	93.3% accuracy	70.3% accuracy

pictorial activities was used to teach living skills to children with ASD [24]. These ATs are not only used for skill development, but also as tools for medical diagnosis and therapy. For example, an assistive alternative communication (AAC) application was used to promote communication between physicians and adult patients with IDD [23].

As ATs have become more widely adopted, several researchers proposed frameworks for designing ATs in learning environments for children with ASD [30, 44]. These design guidelines include using real-world objects, virtual environments, and personalized assistance. Besides, they include experiential learning practices and employing cognitive development models.

2.2 Assistive Wearables for Neurodiverse Users

Recently, researchers began to investigate how wearables assist neurodiverse users. However, most of the work used wearable ATs targeting children with neurodiversity. Prior work focused on assisting with socialization and communication for children with ASD, using head-mounted wearable devices including virtual reality (VR) [8] and augmented reality (AR) glasses [66], as well as body-worn devices such as headbands [25], wrist-worn bracelets [59], chest-worn bands [9], and interactive clothing pieces [31]. In addition to communication and socialization, researchers also aimed at supporting children in other aspects. Several ATs were designed for children with ADHD. For example, an assessment system used wearable sensors to diagnose ADHD [28]. A chest-worn band application was implemented to intervene in impulsive speaking in classes [53]. Moreover, a prototype of a smartwatch application was studied to explore the tensions and challenges related to children's self-regulation [13]. An assistive prototype that captured real-time activities aimed to help users to sustain their attention [55]. Furthermore, concerning head-mounted devices, researchers explored the usage of HoloLens for children with neurodiversity to learn basic daily tasks [2] and the usage of a VR system for storytelling for children with IDD [21].

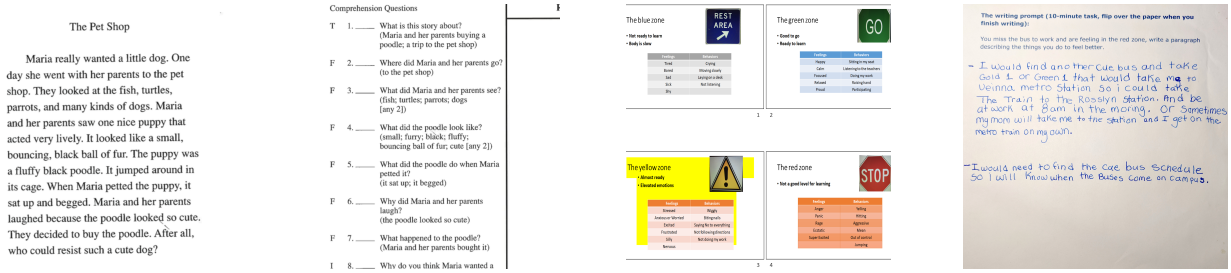
Compared to wearable assistive applications for children, studies involving neurodiverse adults are limited. Previous work investigated how wrist-worn devices (smartwatch and personalized band) helped users to manage anxiety for adults with ADHD and ASD [16, 51]. Also, previous research explored the usage of augmented gloves to provide real-time feedback on work-related activities for employees with cognitive impairments [32]. Still, little is known about the potential opportunities for wearable ATs utilizing AI for young adults with neurodiversity. In inclusive education, it is unclear the potential of datasets to ensure training for machine learning models.

2.3 Wearable and Mobile Sensor Datasets of Human Activities

Wearables and mobile devices have been used extensively to recognize and monitor various human activities in diverse contexts [38]. Previous studies involving data capture in the wild [34, 62, 65] collected data related to fitness and social activities using wearable or mobile sensors.

The dataset generated by Vaizman et al. (2018) tracked users' behavior in activities such as walking, sitting, and laying down, using self-reported context labels [62]. This dataset contributes to the knowledge of human behavior contexts in natural settings, laying a groundwork for in-the-wild data collection studies. Similarly, wearable sensor data was collected to identify hand activities in the study by Laput and Harrison (2019) [34]. Weiss et al. (2016) collected data of fitness activities and hand activities (e.g. eating) from smartwatches' and smartphones' accelerometers and gyroscopes. They compared smartwatch and smartphone-based activity recognition and showed that smartwatches could identify specialized hand activities more effectively than smartphones [67]. Shahmohammadi et al. (2017) used smartwatches to collect real-time data of fitness activities (standing, sitting, walking, and running) from embedded accelerometers and gyroscopes [48]. Then, they used active learning approaches for identifying samples that should be labeled by the user for boosting classification performance. Ni et al. (2019) released a workout dataset comprising sensor data (e.g., heart rate, speed) and contextual data (e.g., longitude, latitude, altitude). They also evaluated a sequential model for classification based on a recurrent neural network (RNN) [42]. Existing benchmarks and datasets represent data collected from users who were adults without any reported disability. Table 1 displays the existing datasets along with their sample size, types of activities, classification models, and results.

To summarize, existing datasets focus mainly on daily whole-body activities, except for the work of Laput and Harrison (2019) whose focus is on hand activities [34]. Also, the datasets for primitive activity recognition focus on locomotion or sports activities [38]. No existing wearable dataset collected data from people with disabilities, which could potentially lead to bias in the trained machine learning models, excluding users with diverse needs [12, 61]. While there are several datasets collected with wearable devices, to the best of our knowledge, there are no hand-activity datasets on learning activities for young adults with neurodiversity. There is a missed opportunity to implement and use assistive wearables that leverage sensor data to harness information and provide individualized support for students with disabilities in inclusive education programs.



(a) Read Material (b) Follow-up Q&A of (a) (c) Slides for Typing Task (d) Prompt Writing Sample

Figure 1: The materials for the learning tasks including (a) the reading material and (b) its follow-up Q&A questions for the easy-level reading and Q&A task, (c) four slides about mood assessment for the typing task, (d) prompt writing topic and a writing sample from a participant.

3 METHODS

To understand the learning behaviors of young adults with neurodiversity, we conducted a user study. The protocol for data collection, the session setup and duration, the smartwatch application implemented for data collection, and the data annotation strategy are thoroughly explained in the following subsections.

3.1 Data Collection

We collected data from the smartwatch built-in sensors regarding four learning activities. Each participant completed the same set of activities, including an easy-level reading with follow-up Q&A, a medium-level reading with follow-up Q&A, copying slides by typing, and prompt writing. The protocol of each session was defined by three special education experts. Each learning activity lasted at most 10 minutes per session, and there was a short break after each session to prevent fatigue. We focus on academic learning activities as these activities are executed across settings (e.g., employment, education, homework) where self-regulation is needed to improve focus and concentration. These activities frequently require the observation and intervention from aides in inclusive/special education programs.

These learning activities were selected after two brainstorming sessions with three special education investigators, including the coordinator of an inclusive higher education program (Mason LIFE) with 18 years of experience who is also a caregiver to a young adult with IDD, and two investigators from the same program with 14 years of special education experience assisting students with IDDs. The activities were decided after discussions involving also an HCI professor, and a PhD student in HCI, both with five years of research on assistive technologies for neurodiverse users. Study participants were recruited from Mason LIFE, an inclusive postsecondary education program that serves 54 students with IDD and employs 100 support staff members.

According to the learning skills of the participants, the coordinator and the special education professor carefully selected materials for all four academic tasks. These included two-level reading and Q&A materials from a book they commonly adopt in special education [29], four slides about mood assessment for the typing task, and a prompt writing topic.

For the two-level reading with follow-up Q&A sessions, the participants read the article assigned and then they wrote their answers to the questions posed, all related to the contents of the article read during the session. For these two sessions, the reading materials and the Q&A writing materials were shown on paper sheets, as illustrated in the Figures 1a and 1b. For the Typing task, the participants typed using a laptop keyboard. As shown in Figure 1c, four slides about how to assess their mood were displayed on another monitor, placed in front of the participant. The participants copied the main text presented in each slide typing the contents. For the Prompt Writing session, the participants wrote an essay about a given topic on a piece of paper. The essay was related to approaches to deal with anxiety when missing a bus (Figure 1d).

3.1.1 Data Collection Procedure. Upon IRB approval, in the Fall 2019, we recruited nine participants with neurodiversity. The data from one participant was later discarded due to an incomplete reading task. For the recruitment of participants, the coordinator of Mason LIFE reached out to the alumni group by email or phone. Participation was voluntary.

To ensure ecological validity and prevent bias from controlled experiments in lab settings, we collected data from participants in a familiar environment, i.e. their house. After a brief introduction about the study and obtaining informed consent, the same instructions were given to all participants by one researcher. We initially introduced all four learning tasks and let participants choose the order of the tasks they wanted to perform. The four learning tasks consisted of: (1) Reading with follow-up Q&A (easy-level), (2) Typing, (3) Prompt Writing, and (4) Reading with follow-up Q&A (medium-level). Before each learning task, the researcher gave in-depth instruction of the particular learning task to the participants. We recorded the data from each participant while they completed a set of four learning tasks.

Each of the eight participants performed the four learning tasks. In total, we conducted 32 learning sessions for the data collection. On average, the participants took approximately 11 minutes to complete an easy-level Reading with follow-up Q&A, 9 minutes for typing tasks, 6 minutes for prompt writing, and 8 minutes for medium-level Reading with follow-up Q&A. We provided flexible break time after each session so that the participants could take longer breaks as needed. Per participant, resting took around four

minutes and a half in total. During data collection, participants were wearing a smartwatch on their dominant hand, i.e. their right hand. The watch ran a wearable app to record sensor data with the respective timestamps. To support the validation of annotations, each session was documented with video recordings, paper field notes, and logs recorded by a customized mobile app implemented by the research team.

3.1.2 Smartwatch Applications for Data Collection. Given customization requirements to implement the app and budget considerations, we chose Wear OS as the platform for the development and Android devices to build our data-collection application. Android watches are more affordable than Apple watches and also more flexible for customized implementations.

We designed and built an Android application on Google Wear OS smartwatch (Fossil Gen 5 Smartwatch) to collect data from built-in sensors including accelerometer, gyroscope, and heart rate sensors. For all the learning tasks, we used data collected from all these sensors as diverse cues for feature representation. To clarify, because the learning activities we collected (i.e., reading and Q&A, writing, and typing) are all hand activities, we intentionally set the sample rate to be high via Android Sensor API. The thread of our watch application collects data from five sensors at a rate of 100Hz per sensor, including raw accelerometer, raw gyroscope, linear acceleration, gravity, and heart rate. The raw accelerometer, linear acceleration, and gravity sensors measure the corresponding acceleration vectors, and the raw gyroscope measures rotation vectors. Besides motion sensors, heart rate sensor measures heartbeats per minute (bpm).

The smartwatch application is designed to have two main functions. One is to record the sensor data, and another is to write the sensor data to CSV (comma-separated value) files, as shown in Figure 2. The smartwatch app has one toggle button for starting or pausing data collection, and one button for writing the sensor data collected to the CSV file. In the study session, the researcher helped the participant to wear the smartwatch and clicked the 'start' toggle button, after that the button switched to a 'pause' option and the screen turned to ambient mode. After the data collection session, the researcher running the experiment selected the 'pause' toggle-button, and then, clicked on the 'write' button to append the data of that session to the corresponding CSV file. Specifically, the 'write' function writes the data from different sensors to separate CSV files. Each data point was written as a row in the CSV file with timestamps in milliseconds and Coordinated Universal Time (UTC) zone. To facilitate the synchronization of timestamps between the video recordings and the data collected from the smartwatch sensors, we also built a mobile application that recorded the starting and ending timestamps in UTC per session in a log file.

3.2 Data Annotation

To annotate the data collected, we relied on observing attentively the video recordings and the data from CSV files generated from the smartwatch application. CSV files include sensor data with timestamps in each row. Besides watching the video recordings, the hand-written field notes and logs from the mobile app were used as complementary materials, mainly to identify the start and end time of each session in order to help the time synchronization between wearable data in the CSV format and the video files. We also carefully discarded the irrelevant parts of the data samples, i.e. we trimmed values collected before and after each activity. For the time synchronization, we used the unified timestamps in UTC from both videos and smartwatch sensor data. To clarify, the sensor data was logged in CSV files with timestamps in UTC. Also, each video has timestamps in UTC.

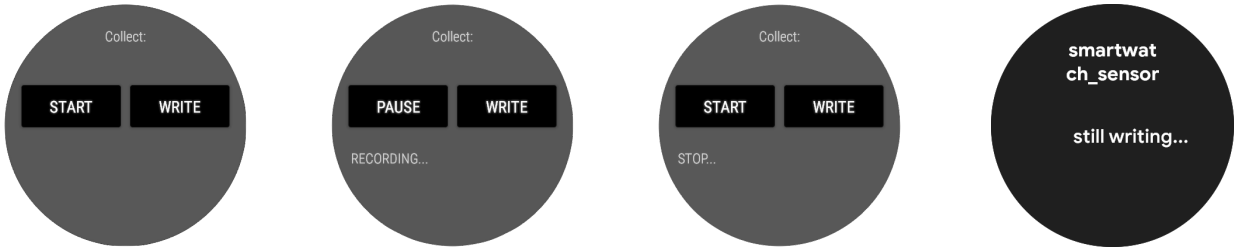
To increase internal reliability, three coders (research assistants) manually annotated the data following a pre-defined agreement on the labeling scheme. They first carefully watched the video recordings and identified six labels (Read, Write, Write Q&A, Type, Off-task, Rest) for each activity based on the participants' behaviors. The labels are described as follows.

- Read: Participants were focused on reading the texts on a pieces of paper.
- Write Q&A: Participants read one question and wrote an answer to it in order. This label is only used in two Reading and Q&A sessions.
- Type: Participants typed on the laptop keyboard.
- Write: Participants wrote on a piece of paper.
- Off-task: Participants lost their attention and did not perform a given learning task for over 2 seconds.
- Rest: Participants took a rest during the break after each learning session.

For the detailed procedure, three coders individually watched and annotated the video recordings by identifying the activities related to the labels listed per session. We observed the change of the learning activities and carefully located the moment when the change of learning behaviors occurred. We identified the activities and located the beginning time and ending time of each activity based on the observed behavior of the participant. Per activity (with a single label), we labeled every second in between the starting and ending time with the corresponding label. For instance, for the Typing session, we identified Type and Off-task. For the Prompt Writing session, we had two activity labels which are Write and Off-task. For the two-level reading and follow-up Q&A sessions, we extracted three kinds of activities including Read, Write Q&A, and Off-task. For the break after each session, we used Rest as a single label. Off-task label is assigned whenever participants lost their focus for more than two seconds in any particular learning task.

This labelling process was continuously performed by three coders to cross-validate the annotated instances for each second. Then, the three coders met to finalize the activity label for each video second and assigned the same label to the corresponding sensor data points at the same timestamp in the CSV files. If more than two coders agreed with one instance, then the labelling of that instance was finalized. If no agreement was reached, the three coders re-watched the corresponding frames of the video to discuss and finalize the labelling of that instance.

For reliability of agreement among three coders' labelling, we calculated the *Fleiss' kappa* reliability of agreement [37]. For details, we treated each second as a case, three raters, and six categories. For *Fleiss' kappa*, the κ is 98.79%, which indicates high agreement level of the annotation.



(a) initial state **(b) start recording** **(c) stop recording** **(d) Store data to files**

Figure 2: Smartwatch application for collecting sensor data with 'start'/'pause' toggle button to control data collection and 'write' button to storage data. To specify, (a) is the initial state, (b) starts collecting data, (c) stops data collection, and (d) is after clicking 'write', in the process of writing sensor data to CSV files.

To summarize the results, we labeled 1,948 seconds as Read; 1,332 seconds as Write; 5,480 seconds as Write Q&A; 3,913 seconds as Type; 1,131 seconds as Off-Task; and 2,112 seconds as Rest. These data labels correspond to the activities collected from eight participants. Detailed results are presented and discussed in the next sections.

4 WLA4ND DATASET

The WLA4ND dataset consists of data collected with a set of wearable sensors, representing learning activities from eight participants with neurodiversity. Among the eight participants, there are three females and five males (according to self-reports), all were alumni from Mason LIFE, the inclusive postsecondary education program for students with intellectual and developmental disabilities. The participants self-reported their disability types. Regarding their demographic profile, seven participants were White-Caucasian, and one was African-American. All eight participants used their right hand as dominant hand. The average age of the participants was 28.5 years old, ranging from 23 to 31 years old. Table 2 lists the details of demographic profile of the eight participants.

The data was collected from five smartwatch built-in sensors, including 13 dimensions of features, being 12 dimensions of features from four motion sensors and one feature from the heart rate sensor. For those four motion sensors that measure vectors in three axes (x, y, z) relative to the smartwatch, we included data from the gyroscope (raw rotations), accelerometer (raw accelerations), linear acceleration (accelerations excluding gravity), and gravity sensors to measure the corresponding acceleration vectors. Heart rate sensors were used to measure the number of beats per minute (bpm) in one vector. For the raw data, around 500 raw time points in total were collected from five sensor sources per second.

Because the difference in hand actions among these learning activities is relatively small, we collected heart rate sensor data to provide more cues apart from motion sensors. For example, 'Rest' and 'Off-task' activities are hard to be distinguished by hand actions. For 'Off-task', the mean beats per minute (bpm) is 63.54 (std = 11.91). For 'Rest', the mean bpm is 73.65 (std = 13.68). However, they are not significantly different by t-test.

The raw sensor signals were pre-processed. The raw time series was chunked by fixed-width sliding windows of 100 data points per window with 50% overlap [3], to generate higher-level feature presentations over this period. Within each sliding window, for each vector per sensor, five metrics were calculated, i.e., the mean, standard deviation, median, 25% percentile, and 75% percentile. These calculations result in 65 features in total. These 65 higher-level features include 60 features (5*3*4) containing five features per axial (x, y, z) dimension from four motion sensor sources, and five features (5*1) from the heart rate sensor collecting one dimension bpm vector. All preprocessing steps were executed using Python scripts. The dataset contains six categories (classes) of labels related to learning activities, which are Read, Write, WriteQ&A, Type, Off-task, and Rest. Considering that all participants read, wrote, typed, and rested while seated, all of these activities are hand activities.

In terms of labels, the pre-processed data includes a total of data points: 3,688 for Read; 2,410 for Write; 1,585 for Write Q&A; 3,533 for Type; 2,385 for Rest; and 610 for Off-task. The total number of data points is 14,211. This results in 9 data points per second.

The WLA4ND Dataset resulted in eight CSV files (one CSV file per participant). Each row of a file has 65 dimensions of features and six columns of six labels in the format of one-hot encoding [26], i.e., a vector representation where its value is 1 and all the remaining elements of the vector are 0. The dataset and the related classification code are publicly available at https://github.com/huizheng-frances/WLA4ND.

5 CLASSIFICATION ON WLA4ND DATASET

The WLA4ND dataset collected from neurodiverse adults comprises learning activities which can be used to develop and train machine learning recognition models. Within the educational setting, such models offer promising support for young adults with neurodiversity in learning activities.

Currently, assistance and interventions are manual. However, this approach can be inconsistent, stigmatizing, and patronizing. With an automated and less intrusive way, assistive wearables relying on activity recognition models can provide discreet interventions on demand. For instance, when an Off-task behavior is detected, the application can gently remind users to take a break. Thus, they can regain focus after a long period of learning activities. Also, the application equipped with automated models could praise users, delivering positive rewards when they concentrate on a task and sustain engagement.

Table 2: Demographic profile of the eight neurodiverse participants recruited in the study.

Participant	Age	Gender	Neurodiversity
P1	24	M	Autism Spectrum Disorder
P2	27	M	Down Syndrome
P3	23	M	ADHD
P5	29	M	Intellectual and Developmental Disability
P6	31	F	ADHD
P7	32	F	Down Syndrome
P8	31	M	Down Syndrome
P9	31	F	Intellectual and Developmental Disability

Given the potential benefits of assistive wearable technologies based on the classification of learning activities, we developed and evaluated five classification models. The models are based on time-series data collected with wearable sensors. We adapted state-of-the-art classification models for time-series data using conventional and deep learning models. We applied both conventional and deep learning activity recognition models including Logistic Regression (LR), Multi-layer Perceptron (MLP), Convolutional Recurrent Neural Network (CRNN) [14] model, Single-layer LSTM (S-LSTM) model, and a Multi-task Hierarchical Attention LSTM (FATHOM) model [11]. The details of the five classification models can be found in Section 5.1. We evaluated these models for User-dependent and User-independent evaluation, as explained in the next subsection.

5.1 Model Summary

Classification Models. We used five state-of-the-art models for time series data. We used Python as a programming language. To build the models, we used Keras neural network library, TensorFlow library, scikit-learn data analysis tool, matplotlib visualization tool, and NumPy library. For user-dependent training, we performed multi-task learning, with one learning task per one participant. Specifically, multiple learning tasks are solved at the same time for all models except for Logistic Regression (LR). For user-independent training, there is only one single task for all models.

- **Logistic Regression (LR)**. LR [46] is a conventional statistical analysis technique employed in machine learning. LR is commonly used across domains. When LR is employed for activity recognition of hand movement, each participant will be responsible for its data for both training and prediction models. LR did not use shared parameters among data of different participants.
- **Multilayer Perceptron (MLP) (16,16)**. MLP (16,16) [63] represents a multilayer perceptron model with two hidden layers in between the input layer and the output layer. In MLP (16,16), the hidden dimensions used for both hidden layers are 16. For multi-task learning, this model learns parameters from multiple participants jointly.
- **Convolutional Recurrent Neural Network (CRNN)**. CRNN [14] is a multi-task learning model that uses Convolutional Neural Networks to extract short-term patterns in a temporal dimension to find local dependencies among features. In this framework, we fed the inputs to a CNN layer and replaced the Recurrent Neural Network (RNN) layer with a single LSTM layer containing 64 hidden cells.
- **Single Long Short Term Memory (S-LSTM)**. LSTM [22] is a technique for sequential data forecasting, known for its capability to capture long-term dependencies. In this model, we used a single layer LSTM. The dimension of the hidden cells is 64.
- **Federated Multi-Task Hierarchical Attention Model (FATHOM)**. FATHOM [11] is a deep learning model that uses attention models and LSTM for time series data. FATHOM uses a hierarchical attention model to learn a common representation from the data. The inter-feature correlations of each user are captured by user-specific attention layers. Meanwhile, the shared temporal correlations are extracted across all users in all time dimensions. Then, FATHOM uses LSTM networks to learn underlying temporal relationships to process the feature representations generated by attention layers. The model is also capable of key feature selections.

6 CLASSIFICATION EVALUATION

For training classification models, we used fixed parameters to generate training models across five models. We used 40 for epochs and 20 for batch size. For evaluation, we report five metrics which are F1 score, Precision, Recall, Accuracy, and the Balanced Accuracy (BA) [63] (designed for imbalanced numbers of labels).

6.1 User-Dependent Evaluation

For the user-dependent evaluation, the classification task was conducted per individual participant. Because the data are time-series data with temporal cues, we used temporal adjacent segments to split the time series data per participant into three parts for cross-validation, including Training data (60%), Validation data (20%), and Test data (20%). The user-dependent classification results among five models are listed in Table 3. To clarify, for user-dependent training, we performed multi-task learning for all models except for LR. Specifically, we calculated the averaged accuracy, BA, F1 score, precision, and recall of the classifications for all participants.

We report that CRNN performs well overall, obtaining the highest measures for four metrics. CRNN shows a 92.2% averaged BA and 83.7% F1 score. FATHOM also performed well with BA (86.0%) and F1 (79.4%). MLP(16,16) also performed relatively well. However, the independently trained LR did not perform as well as the other

Table 3: Results of User-dependent (per-user/P-U) and User-independent (Leave-one-out/L-O) evaluations of five classification models. The evaluation metrics are as follows: BA (averaged Balanced Accuracy), F1 (averaged F1 score), Acc (averaged Accuracy), Pre (averaged Precision), and Rec (averaged Recall).

	BA		*F1*		*Acc*		*Pre*		*Rec*	
Model	P-U (%)	L-O (%)	P-U (%)	L-O (%)	P-U (%)	L-O (%)	P-U (%)	L-O (%)	P-U (%)	L-O (%)
CRNN	**92.2**	81.0	**83.7**	70.0	**94.3**	90.2	81.9	77.1	**86.9**	68.2
FATHOM	86.0	**91.8**	79.4	**91.8**	93.4	**97.5**	**82.6**	98.8	76.6	**86.3**
LR	66.5	50.6	43.9	17.6	80.1	72.5	42.7	19.3	46.0	17.8
MLP(16,16)	83.1	83.6	74.5	82.0	92.9	95.3	81.9	**99.1**	71.4	72.2
S-LSTM	76.8	79.1	63.5	73.4	88.3	92.2	66.1	85.1	61.2	64.9

models. From these results, both deep (e.g. CRNN, FATHOM) and shallow (e.g. MLP(16,16)) neural network models performed better than the regression model (LR). CRNN, FATHOM, and MLP(16,16) outperformed LR in terms of BA by 25.7%, 19.5%, and 16.6% respectively. Figure 3a shows the confusion matrix of the CRNN model. By analyzing the confusion matrix, we noted that the main issue was that the 'WriteQA' samples were misclassified as 'Read' samples. The reason may be because 'WriteQA' is a compound activity that involves reading questions and writing the answers.

Feature Selection. FATHOM seeks to extract feature representations from the input. Also, it learns a common representation with attention mechanisms directly from the data focusing on time dimensions across multiple sensors. Specifically, the attention mechanisms of FATHOM weigh related features corresponding to the sequences of different activities. Based on attention weight matrices of FATHOM, Figure 4 shows the visualization of the attention vector weights across 65 feature dimensions of 150 data points for the 'Read' activity. To clarify, we used only 150 data points of 'Read' activity, which are a small portion of 'Read' activity data. The short sequence (150 data points) cannot represent either the whole 'Read' activities or all the remaining activities. The purpose of this feature selection section is to illustrate the similarities and variations of the selected features from these 150 data points, to show the potential of using the Attention Mechanism [64] for personalization by feature selection.

Figure 4 (a,b) present data from participants P2 and P7, both have Down Syndrome. The features drawn from the accelerometer are common key features on the classification of reading activity for P2 and P7. While P7 has additional key features from gravity and heart rate sensor. Figure 4 (c, d) present the similarities between P3 and P6 (with ADHD), which are different from Figure 4 (a, b). They show that data from the accelerometer, gravity, and linear accelerometer sensors are the most highly related to the reading activities of P3 and P6. While the weight of features from gravity is still slightly higher in P6 than P3.

In summary, although the patterns varied among participants, there are still some similarities. For example, the accelerometer sensor data resulted in a common key feature among four sequences of four participants (as illustrated in Figure 4). Therefore, it is not only important to consider the variations among different kinds of neurodiversity and the varied conditions of individuals, but it is also beneficial to learn notable features related to specific activities.

6.2 User-Independent Evaluation

To investigate the performance of the classification models across users, we conducted a user-independent evaluation. To specify, we applied Leave-One-User-Out analysis for evaluation. More specifically, we performed the leave-one-user-out task 8 times. Each time, we used a different user as the testing set and the data from all other users as the training set. Thus, each user was used once as a testing dataset.

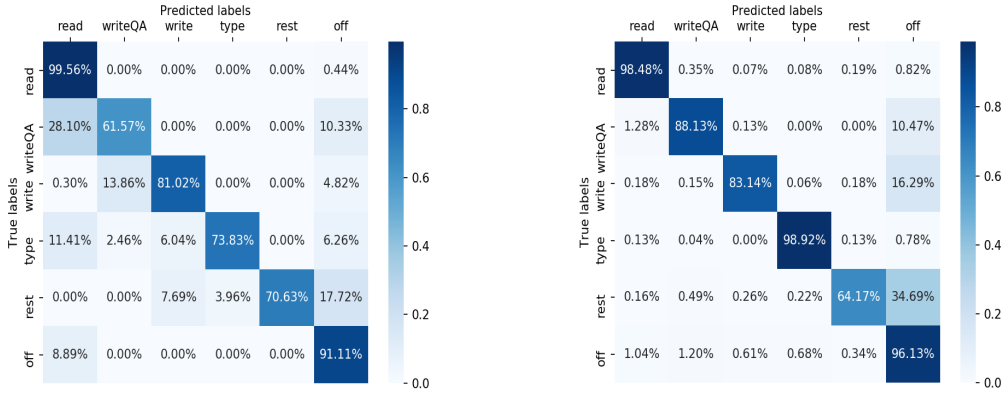
The user-independent classification results (averaged results of the eight experiments) across all users are listed in Table 3. The results indicate that FATHOM performed best, obtaining 91.8% for BA, 91.8% of F1 score, and 97.5% averaged accuracy across all users' data. The CRNN, MLP(16,16), and S-LSTM performed relatively well. Similar to the user-dependent evaluation, LR performed worse when compared to the other models. The confusion matrix (Figure 3b) shows that FATHOM performs very well for leave-One-User-Out evaluation, except in some cases related to the 'Rest' which have been classified as 'Off-task'.

FATHOM performs better in the user-independent evaluation than the user-dependent one given the size of training data. Specifically, for the user-dependent evaluation, per evaluation task, we just used 60% (adjacent time-series segmentation) of the data from a single user for training per user. While, for user-independent evaluation, per evaluation task, we used all the data of the remaining seven users for training which made the training set much larger than in the user-dependent evaluation. Hence, for FATHOM, the user-independent evaluation provides all training data per evaluation task to feed input to this sophisticated model.

6.3 Summary

In a nutshell, through user-dependent and user-independent classifications, the result shows that existing HAR technologies can perform well on the data collected from neurodiverse populations as long as there is a representative dataset for that population.

Overall, deep learning models perform well and better than the conventional method (e.g. LR) on this dataset. To summarize, CRNN performed best (averaged BA of 92.2%) in the user-dependent evaluation, while FATHOM performed best (averaged BA of 91.8%) in the user-independent evaluation among the five models. Meanwhile, FATHOM achieved the second-best performance in the user-dependent evaluation, and MLP (a neural network model) performed the second-best in the user-independent evaluation. Also, S-LSTM performed better than LR in both user-independent and



(a) Confusion matrix of user-dependent evaluation (CRNN) (b) Confusion matrix of user-independent evaluation (FATHOM)

Figure 3: Confusion matrices of (a) user-dependent evaluation (averaged performance for classification evaluations of eight participants) from CRNN and (b) user-independent evaluation (averaged performance of all the eight leave-one-out classification evaluation tasks) from FATHOM.

user-dependent evaluations. For both the user-dependent and user-independent classifications, the confusion matrix shows the possibility of 'Rest' being misclassified as 'Off-task'. For 'Rest', the hand actions are similar to 'Off-task', which may make it hard to distinguish between them. For the feature selection experiment in the user-dependent evaluation, there are both variations and similarities in the key feature patterns among participants. Variations among individuals with neurodiversity as well as common patterns within a specific activity are important.

7 DISCUSSION AND DESIGN IMPLICATIONS

In this section, we reflect on the lessons learned and challenges faced to generate a wearable sensor dataset of learning activities for young adults with neurodiversity. We focus on data collection, data annotation, and classification evaluation. We also provide suggestions and design implications drawing from the research study.

Data Collection in Controlled Settings versus in-the-wild. Collecting data along with self-reported labels in the wild increases the size of the dataset. Despite such benefit, self-reports in the wild would also increase the cognitive load of the participants with neurodiversity during data collection. There are two ways to complete self-reports. One is to send real-time notifications for participants to report activities, and another is to report by recall after finishing the learning task. Real-time self-reports in the wild may interrupt participant's learning activities reducing their concentration on the task. Self-report by recall is less accurate and could be biased, especially for Off-task activities. Besides, some neurodiverse participants present memory limitations which make it harder for them to memorize and recall events accurately. Hence, it leads to a question of how to accurately and efficiently collect more data when involving participants with neurodiversity in research studies. Additionally, we could not collect more data in person to extend the dataset as planned due to the COVID-19 pandemic. One recommendation is to provide a platform that requires caregivers to observe and track the participants' learning activities to get more data with ground truth labels.

Data Annotation. Manual annotation of video recordings is time-consuming, and using online meeting environments for to complete this process remotely adds new challenges to the annotation. Before the quarantine, researchers separately coded each video and met in person to assess reliability and resolve eventual disagreements, re-watching the video and re-labeling the activities based on a common protocol to ensure consistency. Due to the COVID-19, meetings switched to online platforms to complete the annotation process. Specifically, a screen-sharing feature was used to re-watch the videos jointly and to resolve eventual disagreements. To preserve privacy, we never recorded our online meetings and we carefully chose collaboration tools requiring dual-factor authentication, following the university privacy guidelines and IRB policies. We also used encrypted files and secure transmission protocols to exchange data. Multimedia players using top-notch speed control could aid in the process, and teleconferencing tools with real-time synchronous sharing screen function will support distributed annotation sessions. For future work, one potential approach to automate the labeling process is to use the activity recognition models to do the coarse-level annotation first, and the human coders manually refine the generated labels by models. Another approach is to employ crowd-sourcing data annotation with authorized personnel since this process requires considerations of privacy constraints and technical configurations.

Unbalanced Samples. The unbalanced sample noted in the dataset indicates ecological validity by matching the records to real-world situations. Specifically, Read labels are the most frequent because there were two reading sessions among four tasks. Also, the fewer occurrences of Off-task labels are justifiable due to their rare occurrence. We notice the challenge to classify Off-task activities in Figure 3a, where the Off-task instances per user are limited. To overcome this, one recommendation is to use novel one-shot

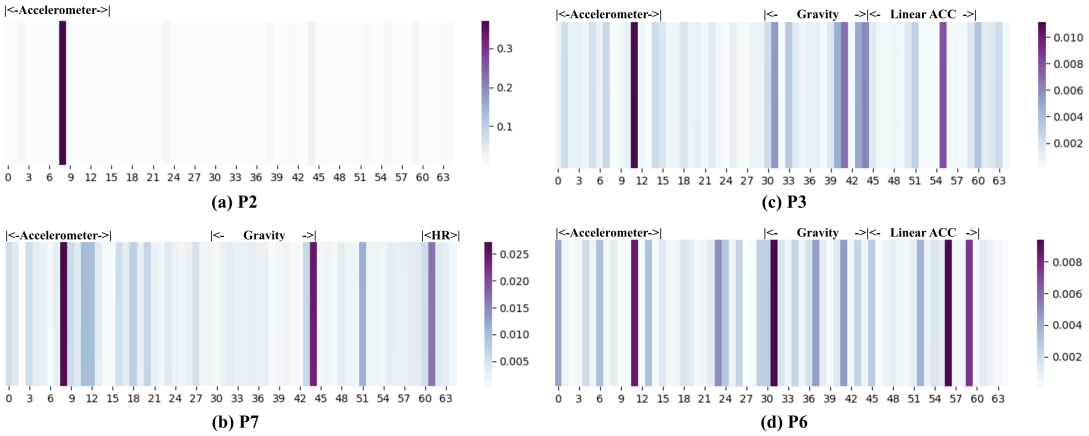


Figure 4: Visualization of selected key features for 150 data points of reading activity, generated via the attention matrix used in FATHOM. The x coordinates represent feature dimensions (65 in total). The legend demonstrates the attention weight implying that the higher the weights are, the darker the color is, and the more important the corresponding sensor-specific features are. The four plots are for four participants (P2, P3, P7, and P6) respectively, where HR stands for Heart Rate sensor and Linear Acc stands for Linear accelerometer. P2(a) and P7(b) have Down Syndrome. P3(c) and P6(d) have ADHD.

learning [17] and few-shot learning [54]. Both approaches aim to learn classification information given only one or a few samples.

In many cases, sampling can mitigate the issues caused by unbalanced sampling. Popular sampling techniques include oversampling and undersampling. Oversampling (e.g., Synthetic Minority Oversampling Technique (SMOTE) [10]) duplicates examples in the minority class or synthesize new instances from the samples in the minority class. Undersampling (e.g., random undersampling) deletes or selects a subset of examples from the majority class [4]. As for our case, data volume and data distribution varied among users. However, the overall distribution is a mixture of all user data distributions. Sampling may lead to a change in the data distribution. Thus, we focused on tuning the machine learning models for better prediction accuracy in this study. We will explore these sampling techniques in future work.

Limitations of Smartwatch Sensors. We observed some technical limitations regarding the smartwatch sensors. Although we set a common frequency among all sensors via Android sensor API, we found that the sampling rate is inconsistent among sensors. Specifically, the sampling rate of the accelerometer is the highest, followed by the gyroscope. Moreover, the sampling frequency of the heart rate sensor is much lower than the motion sensors. Therefore, it took longer for the heart rate sensor to begin data collection. Consequently, it recorded new data points at a slower rate. The sensors with high frequency resulted in fairly dense data points within a given period, while the sensors with lower frequencies resulted in sparse data points. Manipulating Android Wear OS Kernel to stream the accelerometer data at higher frequency [34] is a possible choice. Furthermore, more advanced upcoming hardware devices will likely provide more stable sampling rates. Another limitation of the sensors of the Android smartwatch is that it has just heart rate sensors measuring bpm. To address the lack of physiological data, in addition to the Android smartwatch, other wearable devices can be used to collect data, such as the E4 wristband. This approach implies in additional costs though.

Besides, according to key feature visualizations, applications and models with feature selection can be developed to collect data only from specific sensors, e.g., accelerometer and gyroscope sensors but not necessarily heart rate. Such an approach may also address the inherent limitations of wearable/mobile computing capability and the streaming bandwidth, that facilitates ATs based on ubiquitous computing.

Dataset for Young Adults with Neurodiversity. Although there are mobile and wearable datasets for various human activities from people without disabilities, there is no wearable dataset for learning activities collected from neurodiverse individuals. Without an inclusive dataset collected from diverse users' populations, it is unclear whether the state-of-the-art machine learning models will properly suit these users. Although various AI-based ATs were developed for users with visual/hearing impairments, young adults with neurodiversity are still less supported by ATs utilizing AI due to the lack of inclusive dataset [45]. Despite some limitations (e.g., demographics, types of disability, and population size) in our dataset, it is critical to start a discussion about the possibility of inclusive AI applications to assist young adults with neurodiversity in inclusive learning environments.

8 LIMITATIONS

As discussed in Section 7, due to the hardware limitation of smartwatch sensors, the sampling frequencies among the sensors varied. Besides, there is a limitation regarding the requirement to wear a smartwatch on the dominant hand in this study. Still, there are activities (e.g., typing, off-task) that involve two hands. Also, people may prefer to wear a smartwatch on their non-dominant hand. Additionally, there is a limitation in designing a within-subject experiment because participants needed to perform four learning tasks. Performing more tasks would cause fatigue, therefore, we

limited the number of tasks and did not assign to the participants additional learning tasks (e.g., math calculating).

Although neurodiverse users commonly need assistance to execute learning tasks, the level of execution of the learning activities varies among them. In this study, we did not assess each participant's learning skills. Assessing their specific learning ability (e.g., motor coordination, eye-hand coordination, and cognitive level) and incorporating these differences will be explored as future work. Moreover, due to the difficulty of recruitment and getting the ground-truth data in the studies with controlled settings (rather than using self-reports), the sample size of WLA4ND dataset is relatively small. We plan to collect data with more participants and involve the community to improve the quality of the WLA4NDD dataset.

9 CONTRIBUTIONS

The three main contributions of this paper are stated as follows:

Dataset. WLA4ND dataset was collected from young adults with neurodiversity, a less represented group in Human Activity Recognition (HAR) research. By providing a dataset from populations with neurodiversity, we hope to contribute to inclusive AI technologies. Specifically, WLA4ND can be used to develop and to evaluate activity recognition models, aiming to further facilitate the assistance of people with neurodiversity in inclusive education programs.

Classification. The investigation of classification methods of five models showed that existing HAR technologies can be applied to neurodiverse populations as long as there is a representative dataset collected from that population. It sheds a light to use existing HAR to leverage wearable computing for assistance in inclusive settings. Additionally, we believe that the usage of HAR approaches would alleviate the burden of manual annotation to get more inclusive data with ground truth for people with neurodiversity.

Possible Assistive Applications. An application based on AI models trained on this novel WLA4ND dataset can help young adults with neurodiversity in education by tracking and monitoring their learning activities. Without an inclusive dataset and AI-based ATs, traditionally a support staff must observe the students with neurodiversity while performing academic learning activities, to monitor, track their performance, intervene when needed, and provide positive reinforcement. These assistants are absent when the students are off class or studying remotely. Possible assistive applications will be able to monitor learning activities via HAR. As the interventions to help students to be focused and to positively reinforce them are the most frequent assistance in inclusive education [73], the ATs can use HAR to detect Off-task timing and frequency to provide interventions on demand in order to help students concentrate while studying. Besides, the HAR can be used to monitor the sustained duration of their learning activities to positively reinforce their performance if the duration exceeds a set time threshold. Moreover, by tracking, analyzing, and/or visualizing the monitored learning activity patterns, it will aid the educators and caregivers to know better the students' learning performance to provide tailored programs and suitable study plans for each individual. Moreover, the smartwatch is less-stigmatizing, more affordable, accountable, and consistent (compared to supportive staff). Besides education settings, those applications can be used in training scenarios or inclusive employment, whenever reading, writing, and typing are required.

10 CONCLUSION

This paper presents a novel dataset of learning activities. WLA4ND was collected using built-in smartwatch sensors (accelerometer, gyroscope, and heart rate sensors). The data was collected in 32 learning sessions involving eight participants with neurodiversity, including ASD, ADHD, IDD, and Down Syndrome. The six ground-truth labels used to annotate the dataset are reading, answering follow-up Q&A, prompt writing, typing, being Off-task, and taking a break. Such activities are common in postsecondary education and especially important for assistance in inclusive education. For reliability, WLA4ND was carefully annotated, reaching 98.79% (Fleiss' Kappa) alignment agreement rate among three coders. A classification evaluation using five state-of-the-art machine learning methods was conducted. For the user-dependent evaluation, an averaged balanced accuracy of 92.2% was achieved by CRNN. While, for the user-independent evaluation, 91.8% was achieved using FATHOM which is an attention-based deep learning model. The key features of four short reading sequences were also visualized. The dataset will allow future researchers to explore activity recognition models on it. WLA4ND will pave the way for researchers to implement AI-based assistive technologies (e.g., monitoring, intervention) to support young adults with neurodiversity in inclusive education settings.

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A DATASET INFORMATION

The data collected and in-depth information are described in the WLA4ND dataset ReadMe file https://github.com/huizheng-frances/WLA4ND/tree/main/WLA4ND_dataset. The python code related classification are also available at https://github.com/huizheng-frances/WLA4ND/tree/main/models_code.

B RESEARCH DESIGN

Three figures 5, 6, 7 display the research study environment for the learning tasks.

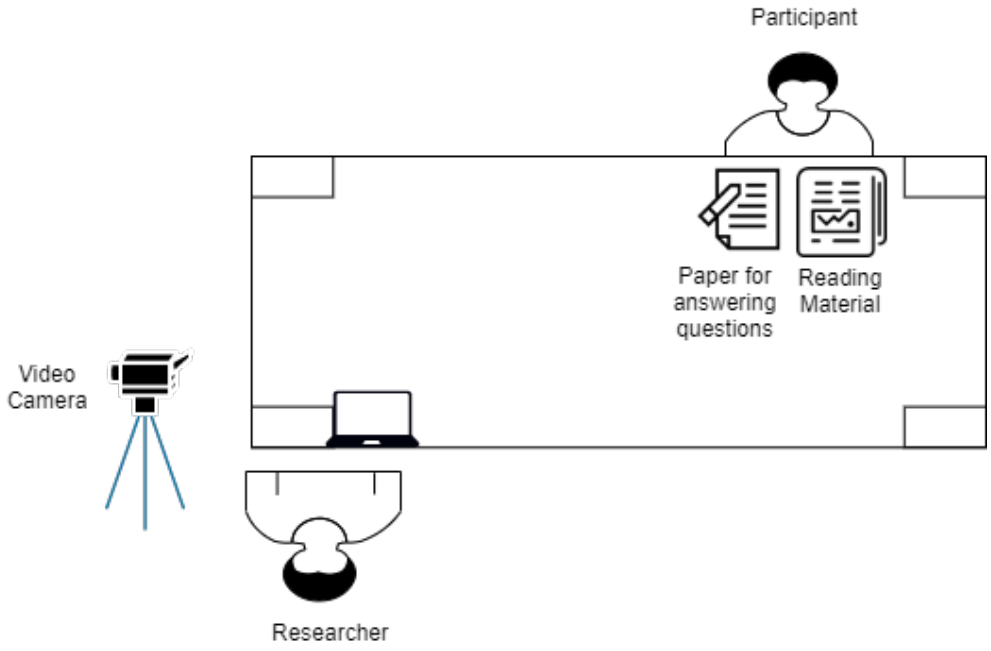


Figure 5: Research design displaying the setup where participants perform easy-level and medium-level Reading with following Q&A task

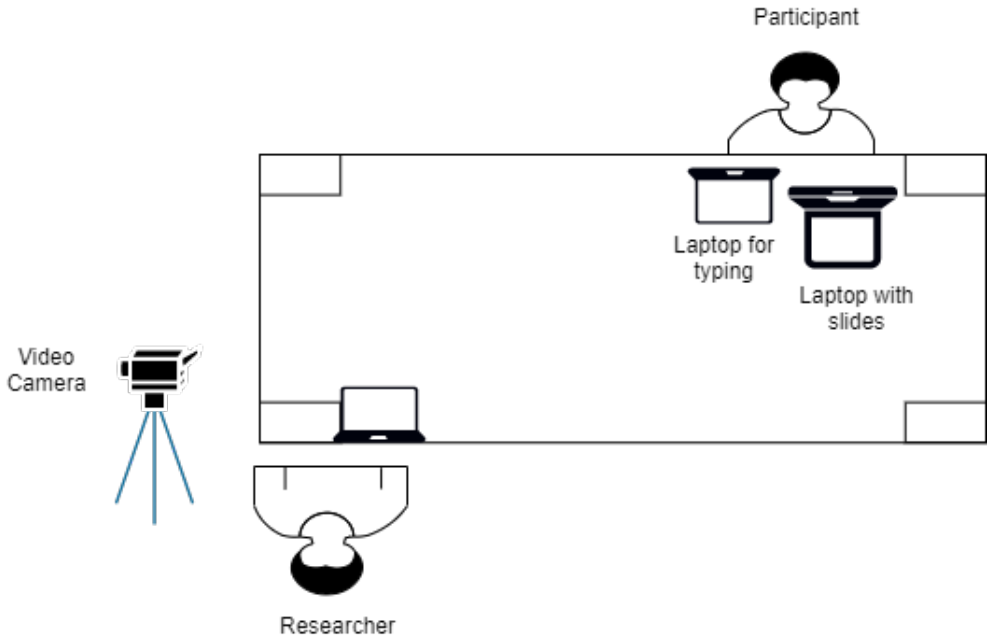


Figure 6: Research design displaying the setup where participants perform Typing task

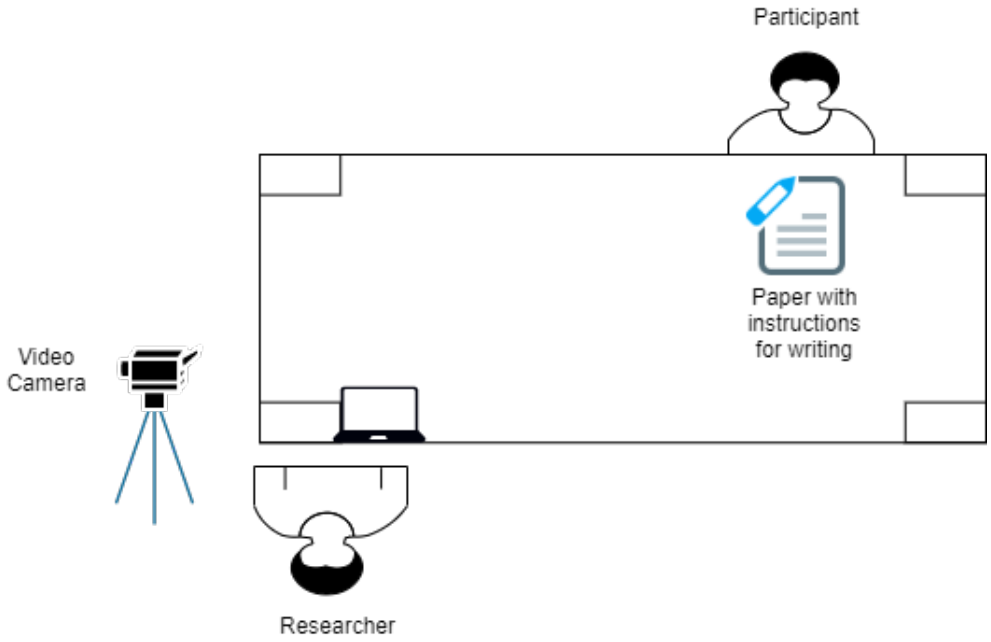


Figure 7: Research design displaying the setup where participants perform Writing task